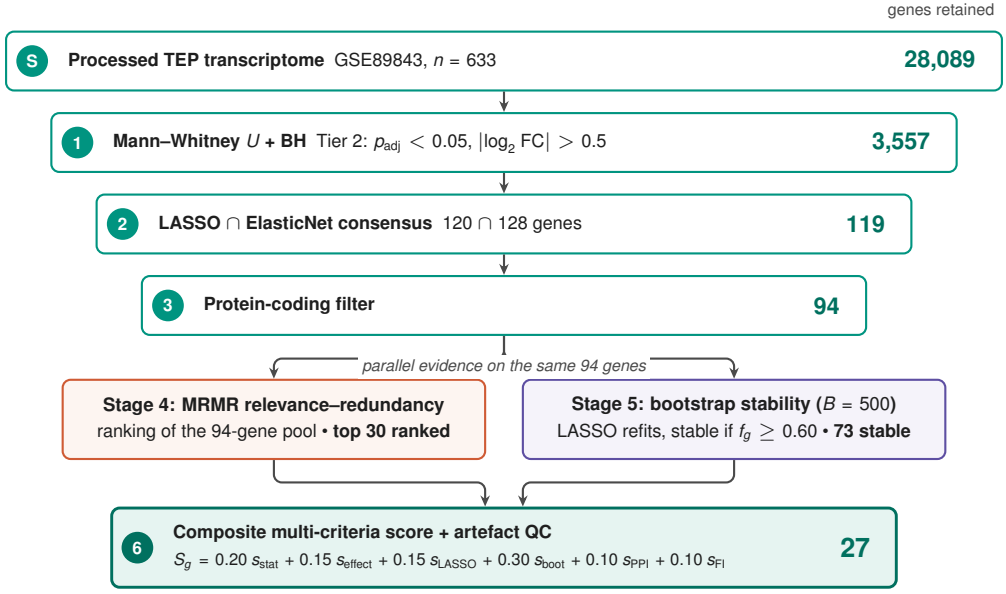

genes retained
S
Processed TEP transcriptome GSE89843, n = 633
28,089
1
Mann–Whitney U + BH Tier 2: p_adj < 0.05, |log2 FC| > 0.5
3,557
2
LASSO ∩ ElasticNet consensus 120 ∩ 128 genes
119
3
Protein-coding filter
94
parallel evidence on the same 94 genes
Stage 4: MRMR relevance–redundancy
ranking of the 94-gene pool • top 30 ranked
Stage 5: bootstrap stability (B = 500)
LASSO refits, stable if f_g ≥ 0.60 • 73 stable
6
Composite multi-criteria score + artefact QC
S_g = 0.20 s_stat + 0.15 s_effect + 0.15 s_LASSO + 0.30 s_boot + 0.10 s_PPI + 0.10 s_FI
27